Computational Physics: Introductory Course

Assignment 4:

Integrals and roots

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What the report should contain

The report should contain:

- The py-files, with carefully commented Python code, that were used to solve the exercises. All code should start with a sentence or two describing what the code is doing. Each py-file should be embedded in the LaTeX or Word document.
- When applicable, you should include the output or plots produced by your code as part of the exercise.
- When you, as part of an exercise, are asked to draw physical conclusions it is important that these conclusions are included in the report. If you are unable to draw any conclusion you should explicitly write 'I was not able to draw any conclusion'.

Important things to keep in mind for producing a good report that adheres to the general rules in scientific writing.

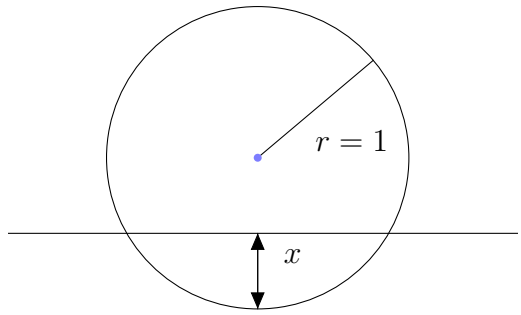
- All variables in the report should be in math font, e.g. the distance was $d = 1.5$ (do not write the distance was d = 1.5, wrong).
- All units in the report should be in correct font, e.g. the acceleration is $a = 1.5$ ms^{-2} (do not write the acceleration is $a = 1.5\,ms^{-2}$, wrong).
- Write $a = 1.5 \cdot 10^{-3}$ instead of $a = 1.5 * 10^{-3}$ (wrong, never use * in equations or text)
- If you give computed values, round them to a reasonable number of decimals.
- The font size in plots should be large enough for the axis texts and legends to be read.

Reading instructions for assignment 4

Start to read chapters 12, 17 and 20 in the lecture notes about functions of one variable, non-linear equations and integrals, respectively.

Exercises

1. A body immersed in a fluid is, according to Archimedes' principle, affected by an upward buoyant force, which equals the weight of the fluid that the body displaces. Use Archimedes' principle to determine how deep a ball of wood with radius 1 m and density $\rho_p = 700$ kg/m^3 floats in water with density $\rho_v = 1000$ kg/m^3. Based on a general physics understanding, start by deciding if x is larger or smaller than 1.



   Hint: the volume of the immersed part of the ball is $V = \pi x \left(x - x^2/3\right)$. Although not mandatory, it would be great if you could prove the formula for the volume of the immersed part.

2. Estimate the definite integral

$$\int_0^3 x^3 \sin(x)\, dx$$

   (a) using the trapezoidal composite rule for $n = 11$, $n = 101$, and $n = 201$ grid points. Is the result dependent on the number of data points? If so, which result is the most accurate? Motivate your answer.

   (b) using the Simpsons composite rule for $n = 11$, $n = 101$, and $n = 201$ grid points. Is the result dependent on the number of data points? If so, which result is the most accurate? Motivate your answer.

   (c) For a given number of grid points, which of the trapezoidal composite rule and Simpsons composite rule gives the most accurate answer? Motivate your answer.

   Hint:

$$\int_0^3 x^3 \sin(x)\, dx = 21 \sin(3) - 9 \cos(3) = 11.87345263\ldots$$

3. The charge distribution of a certain nucleus can be approximated by

$$\rho(r) = \frac{0.084}{1 + e^{(2r-8)}} \quad \left[\frac{\text{q}}{\text{fm}^3}\right],$$

where r in fm (1 fm $= 10^{-15}$ m) is the distance from the center of the nucleus and q is the unit of elementary charge (1 q $\approx 1.602 \cdot 10^{-19}$ C).

(a) Plot the charge distribution of the nucleus. Extend the plotting interval from 0 to the 'practical infinity', a point where the charge distribution is virtually zero.

(b) Using the Simpsons composite rule, estimate the total charge of the nucleus given by the following integral

$$Q = 4\pi \int_0^\infty \rho(r) r^2 dr \quad [\text{q}].$$

Hint: Numerically it is not possible to use infinity as a integration limit and instead you must determine a 'practical infinity', here denoted by r_∞, which is a value of r fulfilling the following conditions:

i. At r_∞, the charge distribution should be virtually zero. Use the plot generated in (a) as a guide.

ii. r_∞ should not be unnecessary large.

iii. Increasing r_∞ should not change the value of the integral to any noticeable extent.

Tip: you can check your result from Simpsons rule by using the built-in command `quad` and setting the upper limit to `np.inf`.

(c) Assuming that the number of protons of the nucleus Z (atomic number), which is an integer, approximately equals the total charge Q, determine what element the charge distribution belongs to (use for example `https://www.webelements.com/`).

Physics background

Archimedes' principle

In *On Floating Bodies*, Archimedes suggested that (c. 246 BC):

> *Any object, totally or partially immersed in a fluid or liquid, is buoyed up by a force equal to the weight of the fluid displaced by the object.*

Archimedes' principle allows the buoyancy of any floating object partially or fully immersed in a fluid to be calculated. The downward force on the object is simply its weight. The upward, or buoyant, force on the object is that stated by Archimedes' principle, above. Thus, the net force on the object is the difference between the magnitudes of the buoyant force and its weight. If this net force is positive, the object rises; if negative, the object sinks; and if zero, the object is neutrally buoyant—that is, it remains in place without either rising or sinking.

Archimedes' principle can be derived by looking at the pressure distribution on an immersed cube, see figure 1.

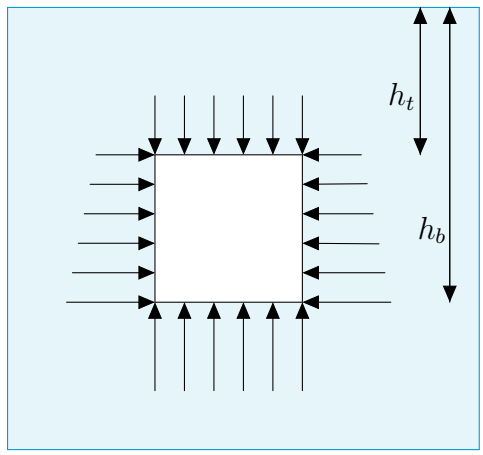


Figure 1: *Pressure distribution on an immersed cube.*

The liquid pressure p is given by

$$p = g\rho h,$$

where ρ is the density of the liquid, g is the gravitational acceleration and h the height of the liquid column or depth within the liquid. The pressures, and thus the forces, on the two vertical sides of the cube exactly balance. The force on the top surface is

$$F_t = g\rho h_t A$$

and the force on the bottom surface is

$$F_b = g\rho h_b A,$$

where A is the top and bottom area of the cube and h_t and h_b are the depths within the liquid of the top and bottom surfaces, respectively. The upward buoyant force is given as the difference

$$F_b - F_t = g\rho \underbrace{A(h_b - h_t)}_{\text{volume of cube}},$$

which is simply the weight of the fluid displaced by the object.

Nuclear charge distribution

The atomic nucleus is the small, dense region at the center of an atom, discovered in 1911 by Ernest Rutherford (Nobel prize 1908) based on the 1909 Geiger–Marsden gold foil experiment, see `https://radioactivity.eu.com/phenomenon/rutherford_experiment` and `https://en.wikipedia.org/wiki/Rutherford_model`. After the discovery of the neutron in 1932, models for a nucleus composed of protons and neutrons were developed. An atom is composed of a positively charged nucleus, with a cloud of negatively charged electrons surrounding it, bound together by electrostatic force. Almost all of the mass of an atom is located in the nucleus, with a very small contribution from the electron cloud. Protons and neutrons are bound together to form a nucleus by the nuclear force.

The diameter of the nucleus is in the range of 1.70 fm, where 1 fm $= 10^{-15}$ m, for hydrogen to about 11.7 fm for uranium. These dimensions are *much* smaller than the diameter of the atom itself, that is in the range of angstroms, where 1 angstrom $= 10^{-10}$ m.

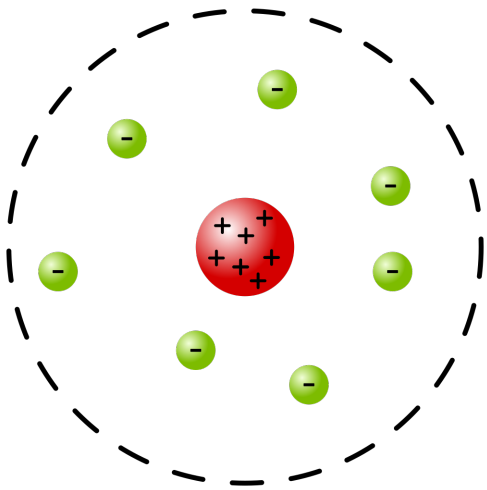


Figure 2: *Rutherfords model of the atom, with the mass concentrated in a small positively charged volume.*

Later electron scattering experiments by Rudolf Hofstadter (Nobel prize 1961) revealed that the nuclear charge distributions $\rho(r)$ approximately follow distributions of the type given in exercise 3. In the assignment on ordinary differential equations later in the course we will theoretically shoot electrons to nuclear targets and determine how the former are deflected.